

WEEK 3 TASK

Unsupervised Learning and Clustering Analysis

K-Means segmentation on the public UCI Wine dataset

Report item	Details
Dataset	UCI Wine dataset (178 observations, 13 chemical features)
Primary method	K-Means clustering with standardized features
Cluster selection	Elbow + Silhouette analysis; k = 3 selected
Visualization	PCA projection, profile heatmap, feature boxplots
Validation	Silhouette score; post-hoc ARI check (K-Means ARI = 0.897)
Software	Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

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1. Executive Summary

This report applies unsupervised learning to the public UCI Wine dataset to identify groups of observations that share similar chemical characteristics. The clustering is performed without using the provided cultivar labels as model inputs. Instead, structure is learned from the 13 measured chemical properties after standardization.

- The dataset contains 178 observations and 13 numeric features; no missing values or duplicate rows were found.
- Feature scaling was necessary because the variables have very different numeric ranges.
- K-Means was selected as the primary algorithm because it is interpretable, efficient for numeric data, and produces centroid profiles.
- Candidate k values from 2 to 6 were evaluated. The highest silhouette score was 0.285 at k = 3.
- The three clusters contain 65, 51, and 62 observations respectively.
- Proline, flavanoids, alcohol, color intensity, and related measures create strong contrasts between cluster profiles.
- The first two PCA components explain 55.4% of standardized variance, so the PCA chart is a visualization aid rather than a full representation of the clustering space.
- A Ward hierarchical model gives a similar silhouette score (0.277), supporting the presence of broad structure beyond one algorithm.

2. Objective and Learning Outcomes

The objective is to demonstrate an end-to-end unsupervised learning workflow: select a suitable public dataset, prepare it appropriately, apply clustering, justify model parameters, visualize the resulting groups, and interpret their characteristics and implications.

- Explain the distinction between unsupervised clustering and supervised classification.
- Understand why scaling and distance metrics matter for K-Means.
- Select a reasonable k using more than one diagnostic.
- Interpret clusters as multidimensional profiles rather than intrinsic labels.
- Use PCA for visualization without confusing a 2D projection with the full feature space.
- Critically evaluate stability, limitations, and practical applications.

3. Dataset Selection and Description

3.1 Dataset source

The analysis uses the Wine dataset distributed with Scikit-learn and originally associated with the UCI Machine Learning Repository. It contains 178 wine observations described by 13 chemical measurements. The three cultivar labels are excluded from model fitting and retained only for optional post-hoc validation.

Attribute	Value
Observations	178
Clustering features	13
Missing values	0
Duplicate rows	0
Known classes	3; not used as clustering inputs

Clustering space	13 standardized dimensions
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3.2 Feature inventory

Feature	Description
alcohol	alcohol
malic_acid	malic acid
ash	ash
alcalinity_of_ash	alcalinity of ash
magnesium	magnesium
total_phenols	total phenols
flavanoids	flavanoids
nonflavanoid_phenols	nonflavanoid phenols
proanthocyanins	proanthocyanins
color_intensity	color intensity
hue	hue
od280/od315_of_diluted_wines	OD280/OD315 of diluted wines
proline	proline

3.3 Why the dataset is appropriate

- Multiple continuous variables make distance-based segmentation meaningful.
- The measurements describe related but distinct chemical properties.
- The sample is small enough for detailed interpretation but large enough for meaningful segmentation.
- Known labels can be used later to check alignment without contaminating the unsupervised fit.

4. Clustering Methodology

4.1 Why K-Means?

K-Means assigns observations to the nearest centroid and repeatedly updates centroids until the assignments stabilize. It was selected because the dataset is numeric, compact, and suitable for interpretable centroid-based grouping. Its main assumption for this exercise is that useful groups can be approximated by compact regions in standardized Euclidean space.

- Advantages: simple, fast, interpretable, and widely used.
- Sensitivity: results depend on k and initialization.
- Distance assumption: Euclidean distance is meaningful after scaling.
- Limitation: K-Means favors compact, approximately spherical clusters.

4.2 Supporting method: Agglomerative Clustering

Ward-linkage Agglomerative Clustering is used as a secondary check. It provides a different clustering mechanism and helps determine whether the broad segmentation is unique to K-Means.

4.3 Analytical pipeline

16. Load the public Wine dataset and isolate the 13 numeric features.

17. Audit missingness, duplicates, and data types.
18. Standardize the features.
19. Evaluate $k = 2$ through 6 using inertia and silhouette score.
20. Fit K-Means with $k = 3$ and a fixed seed.
21. Project the standardized data into two PCA dimensions for visualization only.
22. Profile clusters using raw means and standardized centroids.
23. Compare with a three-cluster Ward model.
24. Interpret evidence, uncertainty, and implications separately.

4.4 Reproducibility settings

```
RANDOM_STATE = 42
N_INIT = 20
K_CANDIDATES = range(2, 7)
SELECTED_K = 3
```

5. Data Preparation for Unsupervised Learning

5.1 Quality audit

The audit found no missing values or duplicate rows, and all clustering features are numeric. The main preprocessing need was scaling.

Check	Result	Action
Missing values	0	No imputation
Duplicate rows	0	No removal
Categorical features	0	No encoding
Scaling	Required	Different units/ranges
Target labels for fitting	Not used	Preserves unsupervised design

5.2 Standardization

Standardization converts each feature to a z-score. This prevents variables with larger numeric ranges, such as proline, from dominating Euclidean distance simply because of their units.

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

6. Choosing the Number of Clusters

The number of clusters is a model-selection decision. Values from 2 to 6 were tested using internal diagnostics instead of assuming k from the source labels.

6.1 Elbow method

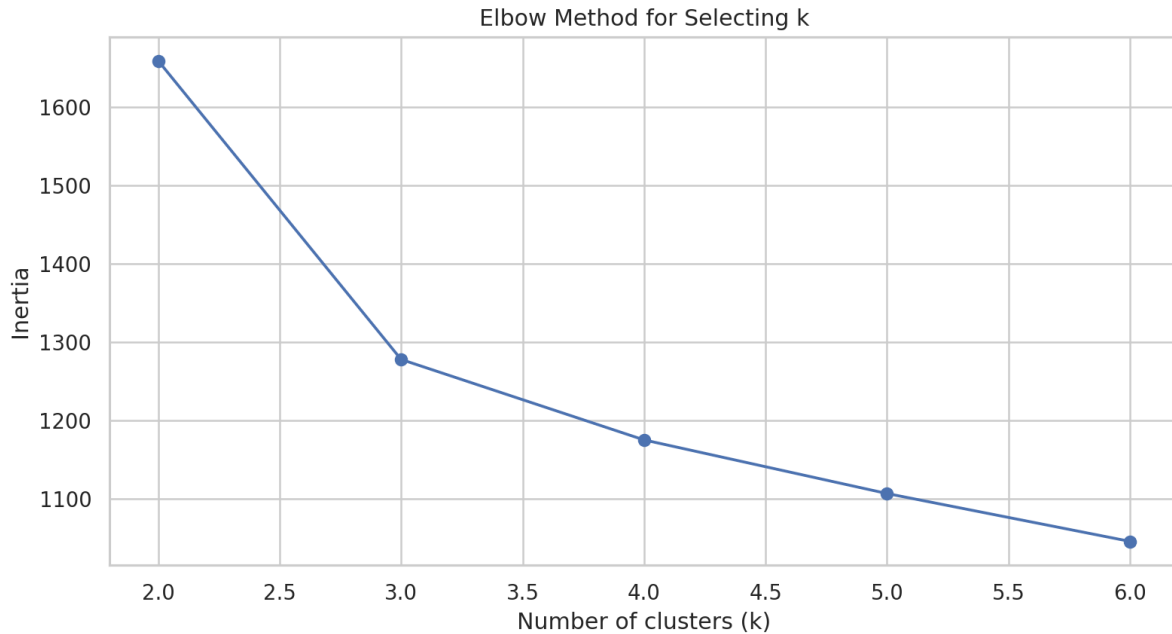


Figure 1. Elbow plot of K-Means inertia across candidate k values.

Inertia decreases as more clusters are added. The curve begins to flatten around $k = 3$ to 4, making $k = 3$ a reasonable simplicity-versus-compactness trade-off.

6.2 Silhouette analysis

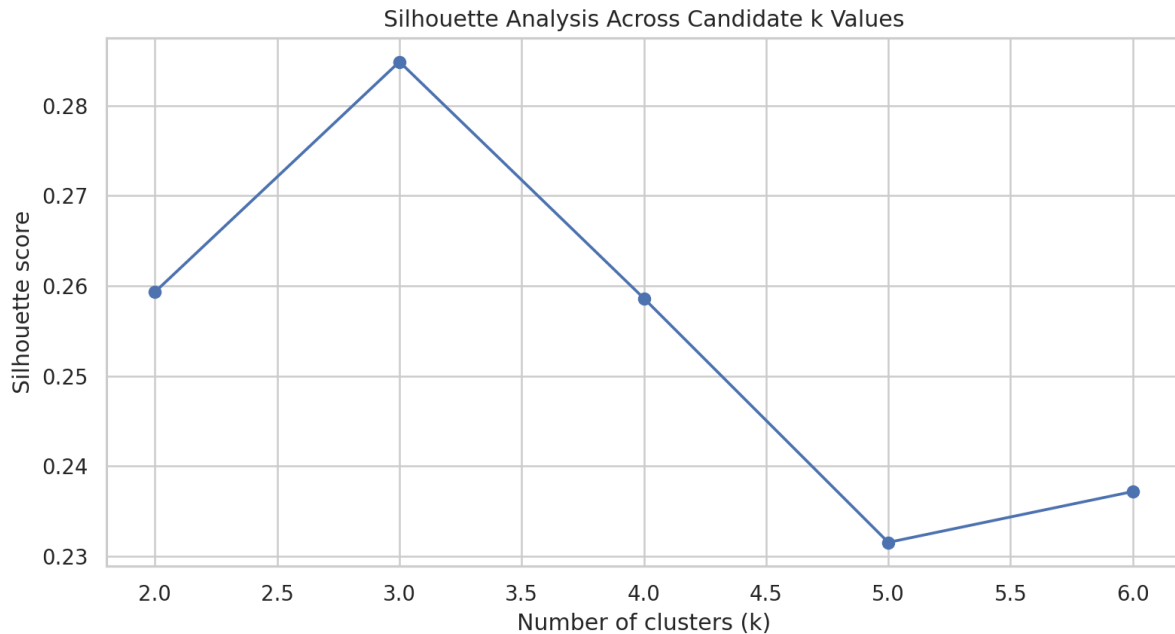


Figure 2. Silhouette scores across candidate k values.

k	Inertia	Silhouette
2	1658.76	0.259
3	1277.93	0.285
4	1175.35	0.259
5	1107.01	0.232
6	1046.00	0.237

The best silhouette score is 0.285 at $k = 3$. Because this is moderate rather than near-perfect separation, the clusters should be treated as informative profiles with some overlap.

6.3 Decision rationale

- $k = 3$ gives the highest silhouette score among tested values.
- The elbow curve has already captured substantial compactness improvement by $k = 3$.
- The resulting groups are reasonably balanced.
- The decision is based only on the unsupervised feature space, not the target labels.

7. K-Means Clustering Results

The final K-Means model uses $k = 3$, `random_state = 42`, and `n_init = 20`. Cluster sizes are 65, 51, and 62 observations.

Cluster	Observations	Share
Cluster 0	65	36.5%
Cluster 1	51	28.7%
Cluster 2	62	34.8%

7.1 Core code

```
kmeans = KMeans(n_clusters=3, random_state=42, n_init=20)
labels = kmeans.fit_predict(X_scaled)
silhouette = silhouette_score(X_scaled, labels)
```

7.2 Overall interpretation

- The groups are relatively balanced rather than being driven by one tiny outlier segment.
- A moderate silhouette score indicates overlapping multidimensional profiles.
- The centroid patterns suggest coherent differences across several chemical variables at the same time.

8. Cluster Visualization and Interpretation

8.1 PCA projection

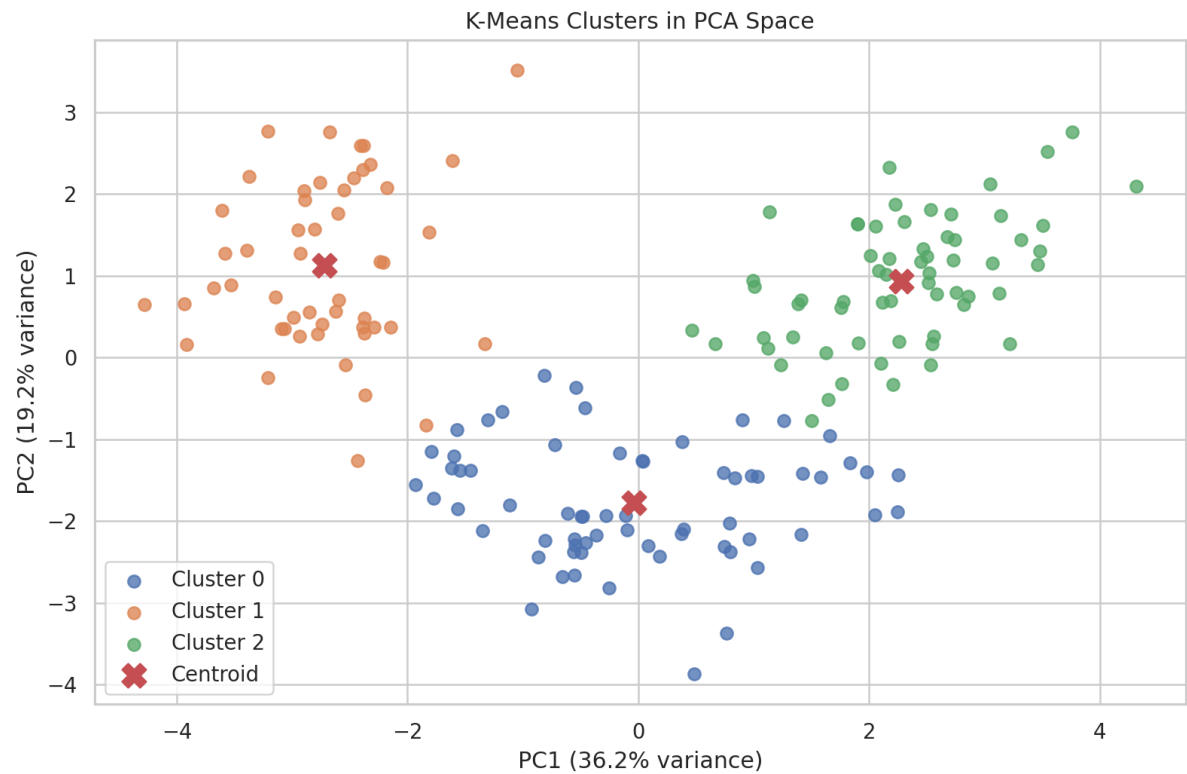


Figure 3. K-Means assignments projected onto the first two principal components.

PC1 explains 36.2% and PC2 explains 19.2% of standardized variance, for 55.4% combined. The clusters occupy partially distinct regions with visible overlap, which agrees with the moderate silhouette score.

8.2 Cluster profile heatmap

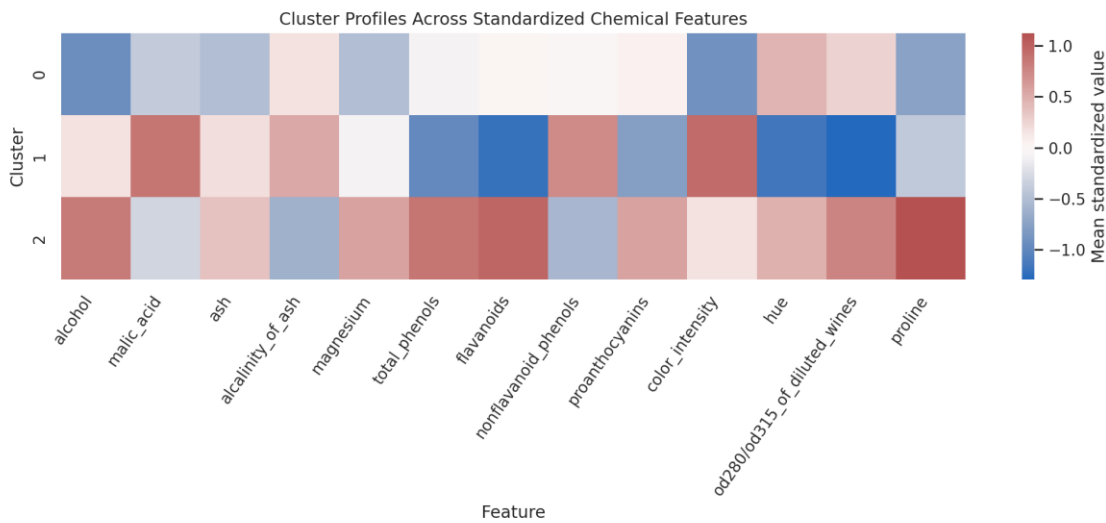


Figure 4. Mean standardized feature values by cluster.

The heatmap compares all 13 variables on a common scale. Cluster 2 shows a strong high-proline and higher-phenolic pattern; Cluster 1 is relatively high in malic acid and lower in hue and OD280/OD315; Cluster 0 is comparatively lower in proline and more moderate across several features.

8.3 Selected feature boxplots

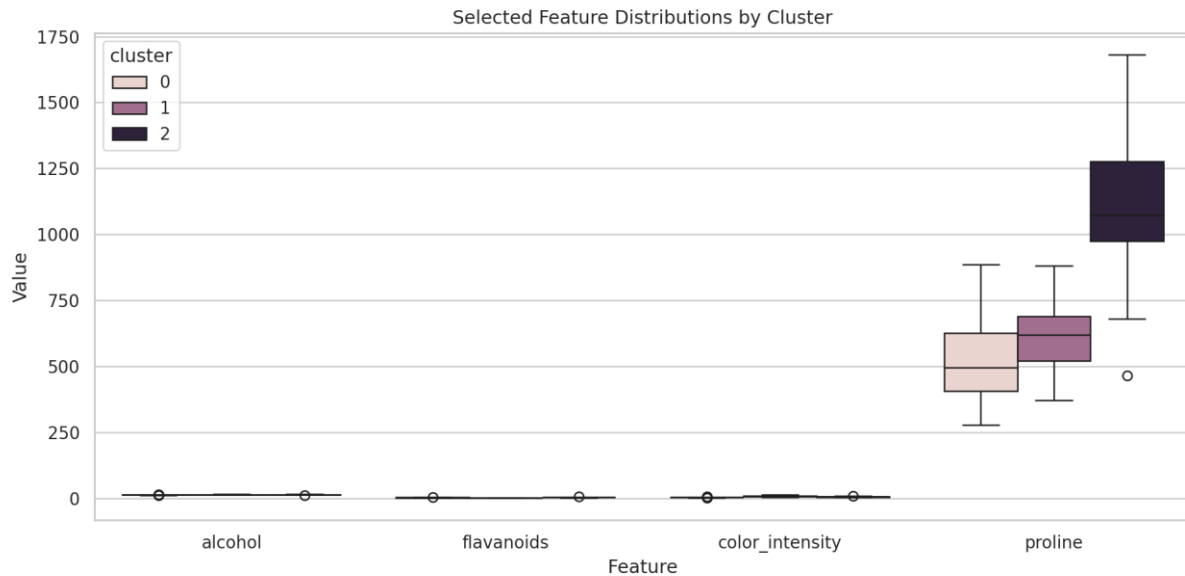


Figure 5. Selected raw-scale feature distributions across clusters.

Proline shows especially strong separation. Alcohol, flavanoids, and color intensity also differ, while overlap remains for some observations, reinforcing the value of a multidimensional approach.

8.4 Most distinguishing features

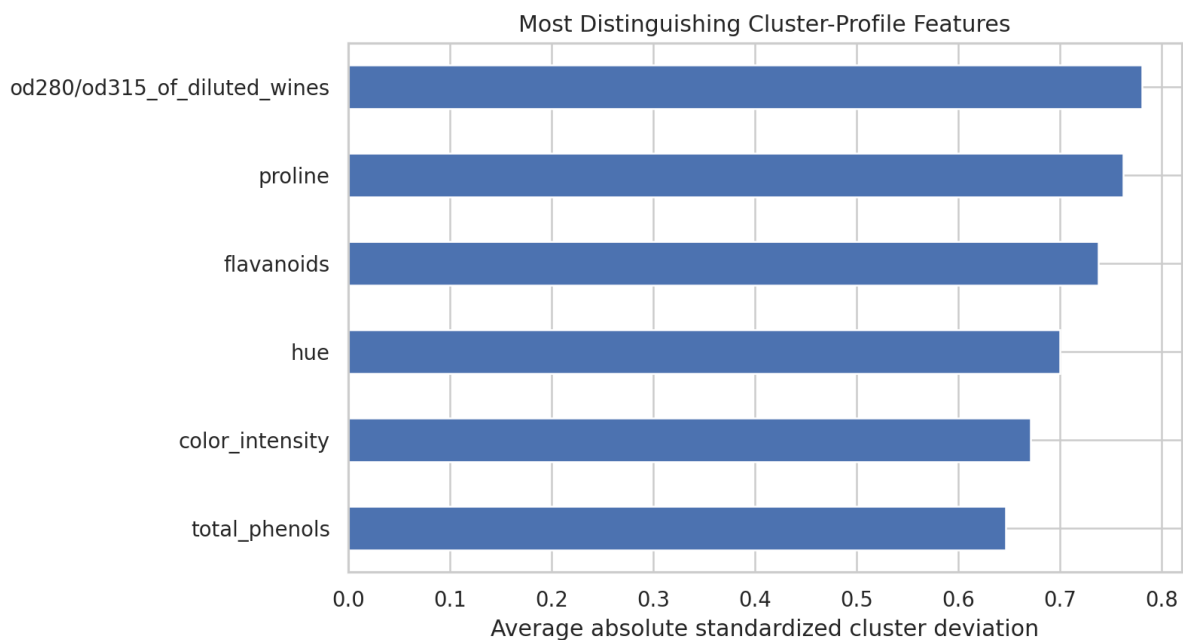


Figure 6. Features with the largest average absolute standardized cluster deviation.

This diagnostic ranks features by how strongly cluster centroids differ from the overall standardized mean. It is an interpretive profile measure, not a predictive feature-importance score.

9. Cluster Profiles and Characteristics

9.1 Cluster-level means

Cluster	N	alcohol	malic_acid	total_phenols	flavanoids	color_intensity	hue	od280/od315_of_diluted_wines	proline
Cluster 0	65	12.25	1.90	2.25	2.05	2.97	1.06	2.80	510.17
Cluster 1	51	13.13	3.31	1.68	0.82	7.23	0.69	1.70	619.06
Cluster 2	62	13.68	2.00	2.85	3.00	5.45	1.07	3.16	1100.23

9.2 Cluster 0 - lower proline / moderate profile

- Contains 65 observations (36.5%).
- Average proline is about 510.
- Average alcohol is 12.25, with moderate flavanoids and OD280/OD315.
- The group is best described as a lower-proline, moderate chemical profile rather than by a single defining variable.

9.3 Cluster 1 - higher malic acid / lower hue and OD ratio

- Contains 51 observations (28.7%).
- Average malic acid is 3.31, the highest of the three.
- Hue is 0.69 and OD280/OD315 is 1.70.
- The cluster combines acidity-related and color/phenolic characteristics.

9.4 Cluster 2 - high proline / strong phenolic signature

- Contains 62 observations (34.8%).
- Average proline is about 1100, clearly above the other clusters.
- Flavanoids are 3.00, alcohol is 13.68, and OD280/OD315 is 3.16.
- This is the clearest high-intensity chemical profile across multiple variables.

9.5 Interpretation caution

Cluster numbers are arbitrary identifiers. They do not imply quality rankings, production tiers, or causal categories. Domain validation is required before operational use.

10. Secondary Validation: Hierarchical Clustering

A Ward-linkage Agglomerative Clustering model with three clusters was fitted on the same standardized feature matrix. It serves as an algorithmic cross-check rather than a replacement for K-Means.

```
agg = AgglomerativeClustering(n_clusters=3, linkage="ward")
agg_labels = agg.fit_predict(X_scaled)
agg_silhouette = silhouette_score(X_scaled, agg_labels)
```

Method	Silhouette	Role
K-Means (k=3)	0.285	Primary
Agglomerative (Ward, k=3)	0.277	Secondary

The K-Means silhouette is 0.285; the Ward model gives 0.277. Their similarity suggests the broad structure is not purely an artifact of one algorithm.

10.1 Optional post-hoc external validation

The original target labels were not used during fitting. After clustering, the adjusted Rand index is 0.897 for K-Means and 0.790 for Ward. These are post-hoc diagnostics, not supervised training scores.

11. Research and Business Implications

11.1 Research implications

- Cluster summaries provide a compact description of multidimensional chemical profiles.
- The profiles can guide hypothesis generation about co-varying measurements.
- Cluster membership can be used in later descriptive analysis if overlap and uncertainty are retained.
- Standardized clusters describe relative, not absolute, chemical levels.

11.2 Business implications

- A winery or quality team could group batches into chemical-profile segments for investigation.
- Cluster profiles could inform differentiated quality-control or sampling workflows.
- Marketing or pricing decisions cannot be justified from this dataset because commercial variables are absent.
- Production use would require stability testing, monitoring, and periodic refitting.

11.3 What the analysis does not establish

- It does not prove that the clusters are objective natural categories.
- It does not establish causality.
- It does not imply that higher measurements are better or worse.
- It does not guarantee the same k or centroids on another wine population.

12. Challenges, Limitations, and Ethical Considerations

12.1 Challenges encountered

- Choosing k without target labels required disciplined use of internal metrics.
- Feature scaling was essential because numeric ranges differ greatly.

- PCA compresses 13 dimensions into two, so visual separation can be misleading if overinterpreted.
- Moderate silhouette scores require cautious language.
- Cluster IDs are arbitrary and meaningful interpretation depends on profiling.

12.2 Method limitations

- K-Means favors compact, approximately spherical clusters.
- Initialization and k can affect the solution.
- Extreme points can influence centroids in other datasets.
- All scaled features are treated as comparably relevant; domain experts may recommend feature selection or transformations.

12.3 Ethical and responsible-use considerations

Clustering can create a false sense of objective categories. Responsible use therefore requires transparent reporting of uncertainty, overlap, assumptions, and validation. Operational decisions should use clusters as decision support rather than as unquestioned truths.

13. Reproducible Python Workflow

The following compact script reproduces the main analysis from dataset loading through model selection, final clustering, visualization preparation, and post-hoc validation.

```
from sklearn.datasets import load_wine
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans, AgglomerativeClustering
from sklearn.metrics import silhouette_score, adjusted_rand_score
from sklearn.decomposition import PCA

wine = load_wine(as_frame=True)
X = wine.data.copy()
y = wine.target.copy() # validation only; not used to fit clusters

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

results = []
for k in range(2, 7):
    model = KMeans(n_clusters=k, random_state=42, n_init=20)
    labels = model.fit_predict(X_scaled)
    results.append((k, model.inertia_, silhouette_score(X_scaled, labels)))

kmeans = KMeans(n_clusters=3, random_state=42, n_init=20)
labels = kmeans.fit_predict(X_scaled)
print('Silhouette:', silhouette_score(X_scaled, labels))

pca = PCA(n_components=2, random_state=42)
X_pca = pca.fit_transform(X_scaled)

profile = (pd.DataFrame(X_scaled, columns=X.columns)
           .assign(cluster=labels)
           .groupby('cluster').mean())

agg = AgglomerativeClustering(n_clusters=3, linkage='ward')
agg_labels = agg.fit_predict(X_scaled)
print('Ward silhouette:', silhouette_score(X_scaled, agg_labels))
print('K-Means ARI:', adjusted_rand_score(y, labels))
```

13.1 Interpretation checklist

- Check feature suitability for the chosen distance metric.
- Scale variables when units/ranges differ.
- Evaluate several k values and more than one diagnostic.
- Inspect cluster sizes for degenerate groups.
- Profile clusters in original units and standardized space.
- Use 2D projections only for visualization.
- Test stability across seeds, algorithms, or preprocessing choices.
- Translate clusters into domain language only after evidence review.

14. Conclusion

This Week 3 analysis applies K-Means to a public, multidimensional wine-chemistry dataset. After standardization, k = 3 was selected from elbow and silhouette diagnostics, producing clusters of sizes 65, 51,

and 62. The cluster profiles differ across multiple chemical variables, with proline, flavanoids, alcohol, and related measurements contributing strongly to the separation.

The analysis demonstrates that successful clustering requires more than executing an algorithm. Data preparation, parameter selection, visualization, profile interpretation, validation, and cautious communication are all essential. The moderate silhouette score and overlapping PCA projection show that the segmentation is useful but not perfectly separated. The hierarchical comparison supports the broad structure while reinforcing that cluster solutions remain model-dependent.

A practical next step would be to evaluate the stability of these segments on new samples and connect cluster membership with operational outcomes before using the results for production or business decisions.

15. References

UCI Machine Learning Repository. Wine Dataset. <https://archive.ics.uci.edu/dataset/109/wine>

Scikit-learn. load_wine. https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_wine.html

Scikit-learn. KMeans. <https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html>

Scikit-learn. silhouette_score. https://scikit-learn.org/stable/modules/generated/sklearn.metrics.silhouette_score.html

Scikit-learn. StandardScaler. <https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html>

Scikit-learn. AgglomerativeClustering. <https://scikit-learn.org/stable/modules/generated/sklearn.cluster.AgglomerativeClustering.html>

Scikit-learn. PCA. <https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html>